Do LLM fingerprints survive outside English?

Interpretable stylometric attribution of five frontier models across a seven-language resource gradient, with a translation extension. Adrian Erlikhman. Repository: github.com/adrian-erlikhman/LangLLM

RQ1	RQ2	RQ3	RQ4	RQ5
59%–72%	**β = -0.038**	**η^2 0.54 vs 0.14**	**ρ = -0.96**	**62% / 63%**
five-way attribution accuracy in every language (chance 20%)	log-odds per resource rank, p = 0.36: no gradient	language vs model; cross-lingual transfer 49%	model separation vs rank: models converge as resources fall	attribution after google / llm translation (English 72%)

Summary

Summary. Five frontier models can be told apart from 21 interpretable, language-comparable features in every one of seven languages, at 59–72% five-way accuracy against 20% chance (logistic regression, leave-one-prompt-out; every 95% CI excludes chance). Contrary to the hypothesis, attribution accuracy does **not** fall along the resource gradient: Japanese is as identifiable as English, Hindi and Turkish sit within a few points of Spanish, and the cell-level gradient coefficient is small and non-significant. Language explains far more feature variance than model does (mean partial η^2 0.54 vs 0.15), yet the model fingerprint is substantially language-invariant: a classifier trained on one language and tested on another, after removing each language's mean, is above chance in all 42 language pairs (mean 0.49). Models **do** converge stylistically as resources fall: the multivariate separation between model centroids shrinks monotonically from English to Hindi (Spearman $\rho = -0.96$, $p < 0.001$), driven mostly by Grok, the outlier in English, drifting toward the pack. The apparent paradox, convergence without loss of attribution, resolves as follows: models converge in the bulk of features while a few high-signal features (paragraph count, comma and colon rates, lexical-diversity slope) keep separating them. Finally, translation does **not** destroy the fingerprint: after Google Translate or a free non-subject LLM renders the English responses into the six other languages, five-way attribution still runs 0.57–0.70, and an English-trained classifier reads the translations at 0.66, because the surviving signal is structural (paragraphing, sentence rhythm, punctuation) rather than lexical. Asked directly which model wrote a text, four of the five models are at chance in every language (GPT names itself 82% of the time regardless of author; the others default to "Claude"); only Claude is above chance (25.4%, $p < 0.001$) with a genuine, if small, self-recognition signal (own-text recall 17% vs 7% false claims) that is strongest in Hindi, not weakest. Interpretable features beat every LLM judge by 35 to 50 points in every language.

1 · Design and method

Subject models (served through OpenRouter; the served model string is logged per response)

model	OpenRouter string
GPT-5.5	openai/gpt-5.5
Gemini 3.5 Flash	google/gemini-3.5-flash
Claude Opus 4.7	anthropic/claude-opus-4.7
Grok 4.3	x-ai/grok-4.3
DeepSeek V4 Pro	deepseek/deepseek-v4-pro-0813

Languages in decreasing order of approximate training-data share (the RQ2 covariate)

rank	language	Stanza/UD package	native length target (in the prompt language)
1	English	en	about 350 words
2	Spanish	es	about 350 words
3	Chinese	zh	about 600 characters
4	Russian	ru	about 350 words
5	Japanese	ja	about 800 characters
6	Turkish	tr	about 350 words
7	Hindi	hi	about 350 words

Cells. 5 models × 12 prompts × 7 languages × 2 generations = 840 responses. Single user turn, no system prompt, temperature 0.7, seeds fixed per generation index, reasoning at low effort with the trace excluded (Gemini 3.5 Flash cannot disable reasoning, so this is the one setting all five accept), 4,000-token budget.

Prompts. Twelve English schemas each fix a topic, a stance (6 for / 6 against), three sub-claims and a reading-level tier (a prompt control only, since Flesch–Kincaid is English-specific). For every language a non-subject model (`meta-llama/llama-4-maverick`) composes a native prompt from the schema, never by translation, so all seven versions share topic, stance and sub-claims and only style is free. A third-family reviewer (`qwen/qwen3.7-plus`) extracted topic, stance and points from each native prompt blind and matched them to the schema; failing prompts were regenerated with the reviewer's note as feedback until all 84 passed. Earlier wordings are retained in each prompt record.

Validation. Language identification (lingua, seven-language closed set, confidence ≥ 0.6), a 150–900 word-equivalent window (Chinese and Japanese converted from characters), a per-language refusal pattern, truncation, and a count of heading or bullet lines.

Features. Every text is parsed with Stanza on Universal Dependencies (tokenize, pos, lemma, depparse), so each of the 21 features has one definition in all seven languages. English-only measures (Flesch, Fog, hedges, passive rate, contractions) are excluded. Chinese and Japanese are pre-split on their terminal punctuation because Stanza's splitter does not handle it.

The 21 features

group	features
Lexical	MATTR (window 50); hapax rate; mean token length; Zipf slope
Syntactic	sentence length mean and SD; burstiness; dependency depth; subordinate-clause rate (acl, advcl, ccomp, xcomp, csubj); function-word ratio; first-person rate (Person=1, lexical fallback for zh/ja)
Structure	paragraph count; paragraph length; question rate; connective rate (cc, mark, sentence-initial ADV/CCONJ/SCONJ)
Punctuation	comma, colon, dash, semicolon per 1 000 tokens, with full-width and ideographic script equivalents mapped

group	features
Character	character-bigram entropy; digit rate (Unicode Nd per 1 000 characters)

Analysis

- **RQ1** — per-language five-way attribution, leave-one-prompt-out (both generations of a prompt stay on one side of the split). Logistic regression on standardised features is the interpretable primary; a random forest is the ceiling. Chance = 0.20; 95% bootstrap CIs; exact binomial tests.
- **RQ2** — Spearman ρ over the seven languages and a cell-level binomial GLM correct ~ rank with prompt-clustered standard errors. Robustness: features residualised on log length within language.
- **RQ3** — per feature, two-way ANOVA (model × language, prompt as blocking factor) with partial η^2; and a cross-lingual transfer matrix of within-language z-scored classifiers.
- **RQ4** — within-language model separation after pooled z-scoring: mean pairwise centroid distance, silhouette of model labels, between/within scatter ratio; stratified reflected bootstrap CIs; Spearman against rank.
- **RQ5** (extension suggested by Philo) — every kept English response translated into the six other languages by Google Translate and by a free non-subject, non-Gemini LLM (MiniMax M3; four fallbacks to Llama 4 Maverick). T1: attribution on translated text alone. T2: classifiers trained on native responses in the target language, and on the English originals, applied to the translations. T3: per-feature Spearman ρ between original and translation.
- Pre-specified decision rules are in docs/analysis_plan.md; RQ5 was added to the plan before any translated data existed.

2 · Data quality

839 of 840 responses are used. Ten DeepSeek cells spent an entire 4 000-token budget on hidden reasoning and returned nothing; they were re-collected at 16 000 tokens, with the override recorded per response. One Grok response to a Hindi prompt came back in English and is excluded. Median hidden reasoning tokens per response: Claude 0, GPT 23, Grok 489, DeepSeek 552, Gemini 1 122.

Keep rate per model × language (after the re-collection)

model	en	es	zh	ru	ja	tr	hi
Claude Opus 4.7	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DeepSeek V4 Pro	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Gemini 3.5 Flash	1.00	1.00	1.00	1.00	1.00	1.00	1.00
GPT-5.5	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Grok 4.3	1.00	1.00	1.00	1.00	1.00	1.00	0.96

Median response length in word-equivalents (Chinese and Japanese converted from characters)

model	en	es	zh	ru	ja	tr	hi
Claude Opus 4.7	354	366	574	338	382	320	382
DeepSeek V4 Pro	354	373	471	324	406	322	386
Gemini 3.5 Flash	333	352	433	308	334	292	390
GPT-5.5	414	414	586	372	396	330	406
Grok 4.3	298	326	387	254	262	226	262

Median Stanza tokens per response: GPT-5.5 is the longest and Grok the shortest in every language

model	en	es	zh	ru	ja	tr	hi
Claude Opus 4.7	362	372	560	332	518	338	381
DeepSeek V4 Pro	361	379	468	322	540	332	386
Gemini 3.5 Flash	346	360	422	308	447	308	390
GPT-5.5	422	426	555	370	524	346	406
Grok 4.3	306	332	368	255	342	236	263

3 · RQ1 — attribution within each language

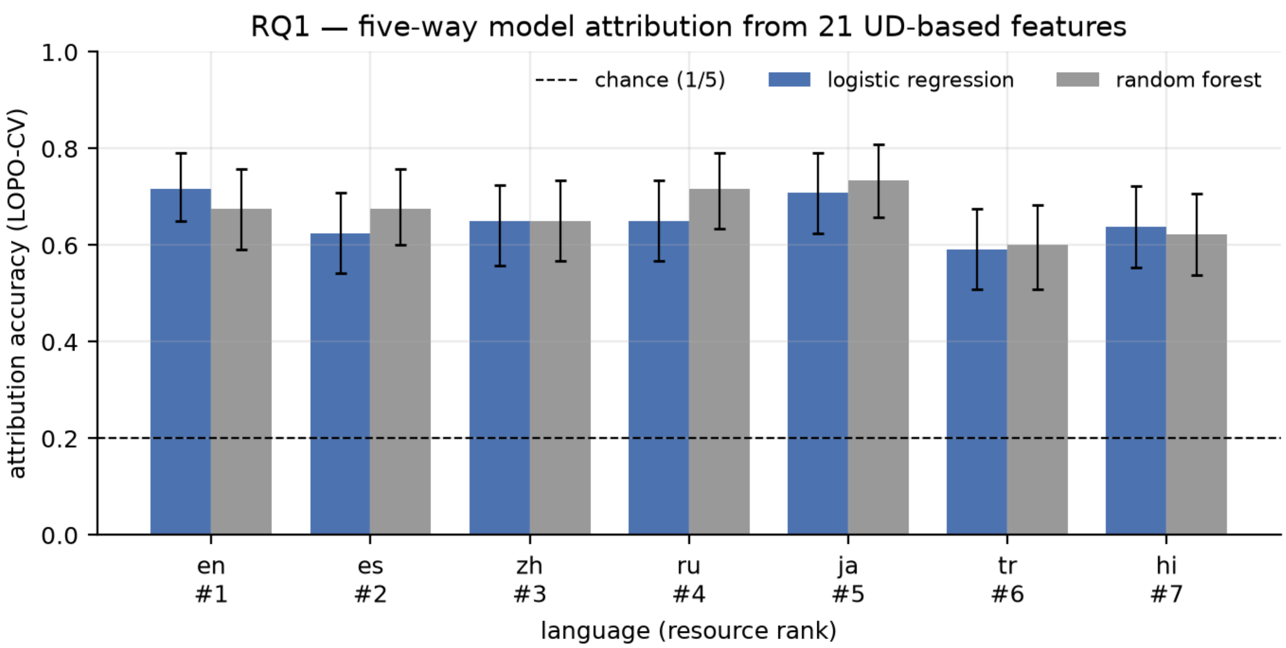

Figure 1. Five-way attribution accuracy per language, leave-one-prompt-out, 95% bootstrap CIs. Dashed line: chance (0.20).

Table 3.1. Attribution accuracy (every binomial test against chance has $p < 0.001$)

language	rank	n	accuracy (logreg)	CI lo	CI hi	macro-F1	random forest
English	1	120	0.717	0.650	0.792	0.711	0.675
Spanish	2	120	0.625	0.542	0.708	0.625	0.675
Chinese	3	120	0.650	0.558	0.725	0.644	0.650
Russian	4	120	0.650	0.567	0.733	0.654	0.717
Japanese	5	120	0.708	0.625	0.792	0.707	0.733
Turkish	6	120	0.592	0.508	0.675	0.596	0.600
Hindi	7	119	0.639	0.554	0.723	0.634	0.622

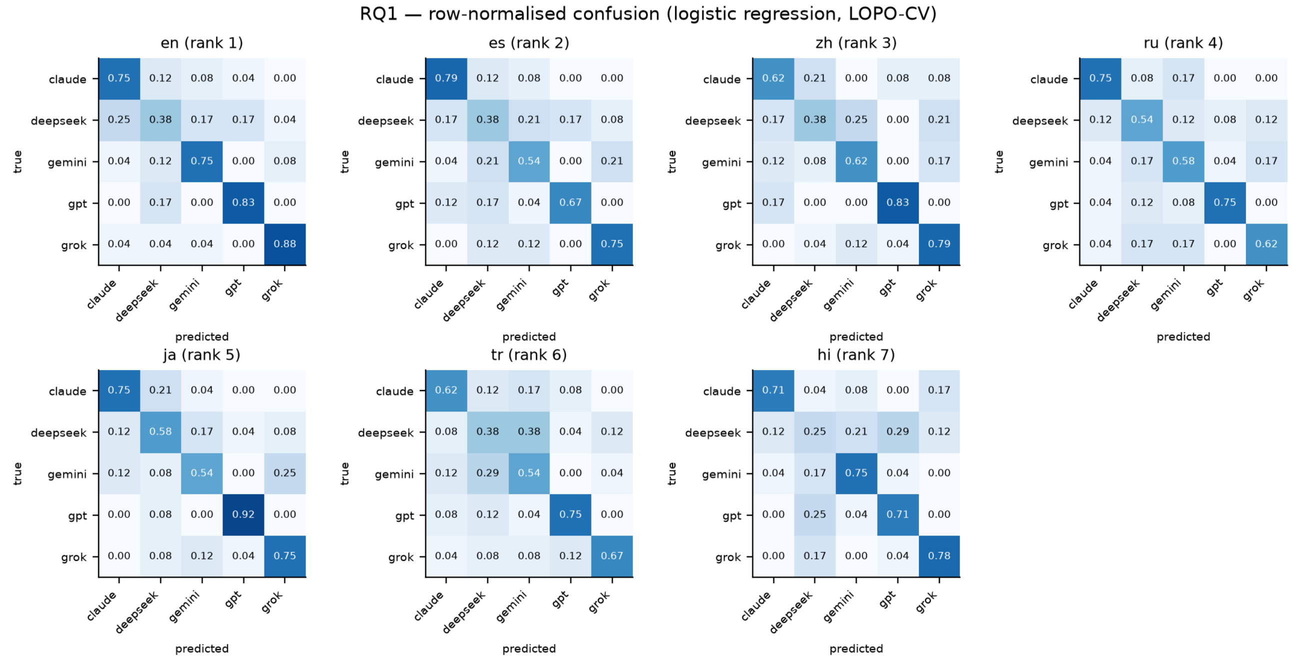

Figure 2. Row-normalised confusion matrices, logistic regression, leave-one-prompt-out.

Table 3.2. Most discriminative features per language

language	five most discriminative features (mean |standardised coefficient| over folds)
English	para_len_mean, subord_rate, func_word_ratio, zipf_slope, mean_token_len
Spanish	colon_per_1k, comma_per_1k, first_person_rate, para_len_mean, para_count
Chinese	bigram_entropy, mattr, comma_per_1k, para_count, semicolon_per_1k
Russian	para_count, comma_per_1k, colon_per_1k, para_len_mean, dash_per_1k
Japanese	bigram_entropy, para_count, comma_per_1k, para_len_mean, zipf_slope
Turkish	para_len_mean, para_count, connective_rate, mattr, bigram_entropy
Hindi	comma_per_1k, bigram_entropy, para_count, zipf_slope, digit_rate

Interpretation. Attribution works in every language. GPT-5.5 and Grok 4.3 are the most identifiable (typically 75–90% recall); Claude Opus 4.7 and Gemini 3.5 Flash follow; DeepSeek V4 Pro is the blur, at 25–58% recall, most often mistaken for Claude in English and for GPT or Gemini in Hindi. The feature-importance tables show the fingerprint is carried by *structure and punctuation* more than syntax: paragraph count and length, comma and colon density, and the lexical-diversity measures (hapax rate, Zipf slope, bigram entropy) dominate in almost every language, whereas dependency depth, subordination and question rate contribute little. Length is part of the signal: GPT-5.5 writes the longest responses and Grok the shortest in all seven languages (§2), and the length-residualised robustness run (§4) shows the *linear* fingerprint loses 10–20 points without it, while the random forest does not, so the residual, non-length fingerprint is real but non-linear.

4 · RQ2 — the resource gradient

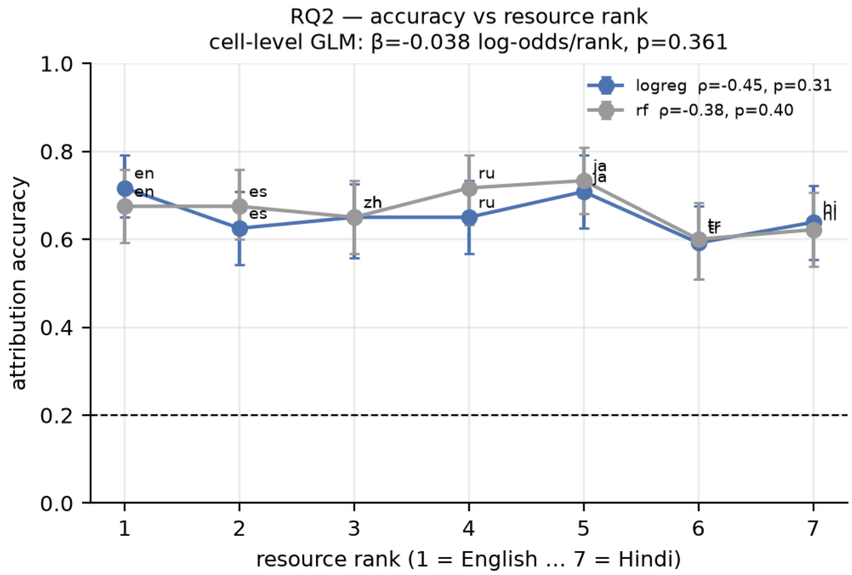

Figure 3. Attribution accuracy against resource rank (1 = English … 7 = Hindi).

Table 4.1. Gradient tests

test	estimate	p
Spearman ρ, accuracy vs rank (logreg, n = 7)	-0.45	0.310
Spearman ρ (random forest)	-0.38	0.403
OLS slope, accuracy per rank step	-0.009	0.348
Cell-level GLM β(rank), log-odds, n = 839, prompt-clustered SE	-0.038 (SE 0.042)	0.361
GLM β(rank) with length-residualised features	-0.053	0.164

Table 4.2. Robustness: each feature residualised on log length within language

language	logreg, length-residualised	random forest, length-residualised
English	0.567	0.658
Spanish	0.483	0.775
Chinese	0.442	0.683
Russian	0.467	0.750
Japanese	0.500	0.750
Turkish	0.450	0.708
Hindi	0.445	0.664

Interpretation. The pre-registered decision rule (negative rank coefficient, $p < 0.05$) is **not met**. The point estimate is negative but tiny: about −0.04 log-odds per rank step, which is under one percentage point of accuracy per step, with $p \approx 0.36$ ($p \approx 0.16$ after length control). The ordering is also not monotone: Japanese (rank 5) matches English, Hindi (rank 7) beats Turkish (rank 6) and Spanish (rank 2). With interpretable features, the fingerprint does not fade with training-data share; whatever these models do differently, they do it in Hindi too. The one qualification is that our rank is ordinal and approximate; a numeric share covariate might sharpen the picture, but the range of accuracies (0.59–0.72) leaves little gradient to find.

5 · RQ3 — model style vs language style

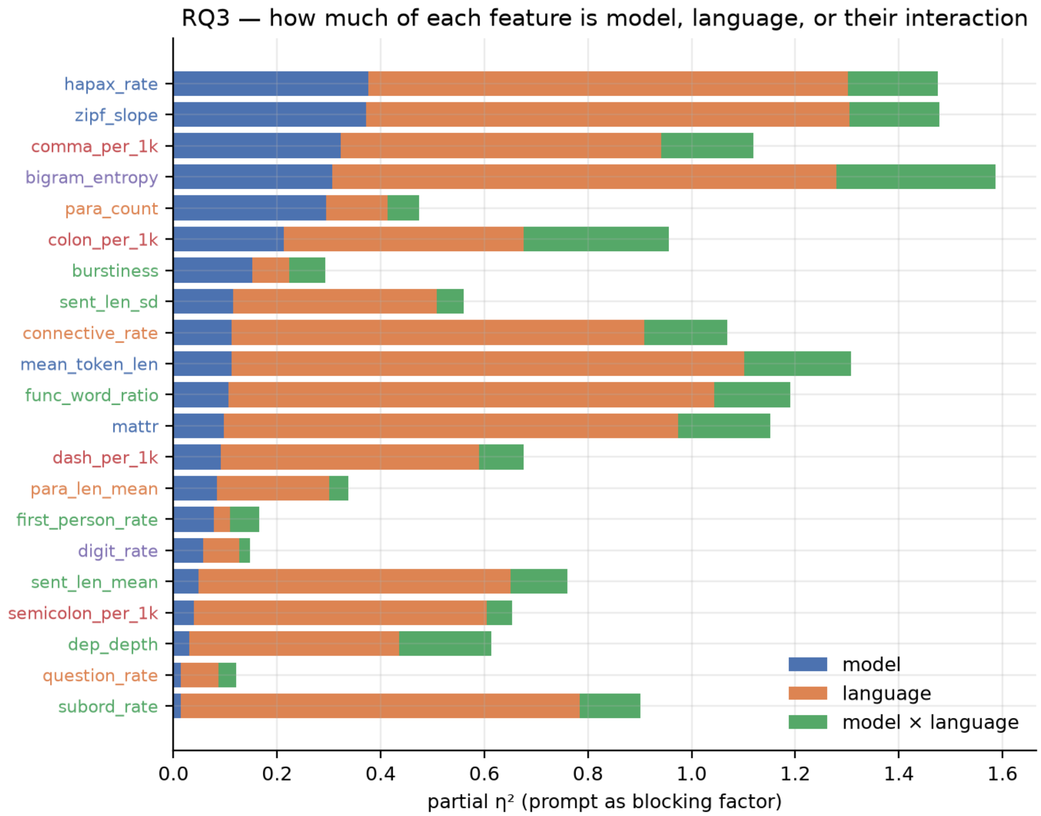

Figure 4. Partial η² per feature for model, language and their interaction (prompt as blocking factor). Label colour = feature group.

Table 5.1. Partial η² per feature, sorted by model effect. Mean over 21 features: model 0.145, language 0.539, interaction 0.127.

feature	η² model	η² language	η² model×language	η² prompt	p model	p interaction
hapax_rate	0.376	0.927	0.173	0.257	0.000	0.000
zipf_slope	0.372	0.933	0.174	0.223	0.000	0.000
comma_per_1k	0.323	0.619	0.177	0.199	0.000	0.000
bigram_entropy	0.306	0.974	0.306	0.120	0.000	0.000
para_count	0.294	0.119	0.061	0.100	0.000	0.001
colon_per_1k	0.213	0.463	0.280	0.028	0.000	0.000
burstiness	0.152	0.071	0.070	0.078	0.000	0.000
sent_len_sd	0.116	0.392	0.052	0.174	0.000	0.011
connective_rate	0.112	0.796	0.161	0.193	0.000	0.000
mean_token_len	0.112	0.990	0.206	0.420	0.000	0.000
func_word_ratio	0.107	0.937	0.147	0.263	0.000	0.000
mattr	0.098	0.877	0.178	0.150	0.000	0.000
dash_per_1k	0.091	0.498	0.086	0.067	0.000	0.000
para_len_mean	0.084	0.216	0.038	0.090	0.000	0.161
first_person_rate	0.078	0.031	0.057	0.126	0.000	0.003
digit_rate	0.057	0.070	0.020	0.277	0.000	0.879
sent_len_mean	0.049	0.601	0.110	0.151	0.000	0.000

feature	η² model	η² language	η² model×language	η² prompt	p model	p interaction
semicolon_per_1k	0.040	0.564	0.049	0.079	0.000	0.019
dep_depth	0.030	0.406	0.178	0.083	0.000	0.000
question_rate	0.014	0.073	0.034	0.107	0.025	0.257
subord_rate	0.014	0.770	0.117	0.159	0.029	0.000

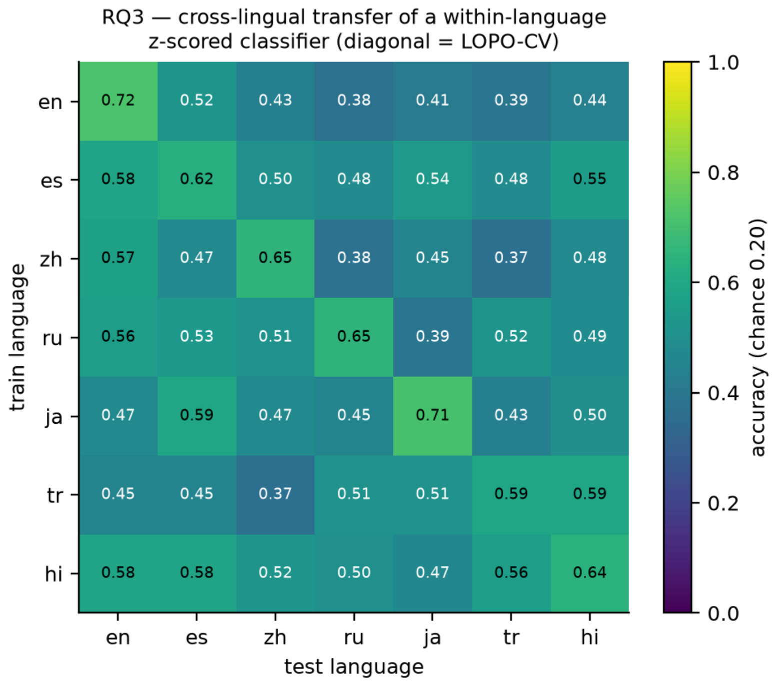

Figure 5. Cross-lingual transfer of a within-language z-scored classifier (diagonal = leave-one-prompt-out).

Table 5.2. Transfer accuracy, rows = training language, columns = test language. Mean off-diagonal 0.486 vs diagonal 0.654; 42/42 off-diagonal cells above chance.

train \ test	en	es	zh	ru	ja	tr	hi
en	0.72	0.52	0.43	0.38	0.41	0.39	0.44
es	0.58	0.62	0.50	0.48	0.54	0.48	0.55
zh	0.57	0.47	0.65	0.38	0.45	0.37	0.48
ru	0.56	0.53	0.51	0.65	0.39	0.52	0.49
ja	0.47	0.59	0.47	0.45	0.71	0.43	0.50
tr	0.45	0.45	0.37	0.51	0.51	0.59	0.59
hi	0.58	0.58	0.52	0.50	0.47	0.56	0.64

Interpretation. Language dominates every feature that touches morphology or the lexicon: mean token length, function-word ratio, MATTR, hapax rate and Zipf slope have language η^2 above 0.85, as expected when comparing Chinese characters, Turkish agglutination and Hindi postpositions. Model effects are largest on the same lexical-diversity measures (hapax rate 0.38, Zipf slope 0.37), on comma density (0.32), bigram entropy (0.31) and paragraph count (0.29), and smallest on subordination, question rate and dependency depth (≤ 0.03), so the models differ in *how varied their vocabulary is and how they punctuate and paragraph*, not in clause architecture. The interaction term is on average as large as the model term (0.13 vs 0.15), meaning each model adapts its style per language, but the transfer matrix shows the invariant part is the larger share: after within-language standardisation, a classifier trained on any language reaches 0.38–0.59 on any other (mean 0.49 against a within-language mean of 0.65; all 42 off-diagonal cells above chance). Roughly three quarters of within-language performance transfers. The fingerprint is therefore mostly language-invariant with a per-language accent.

6 · RQ4 — convergence in low-resource languages

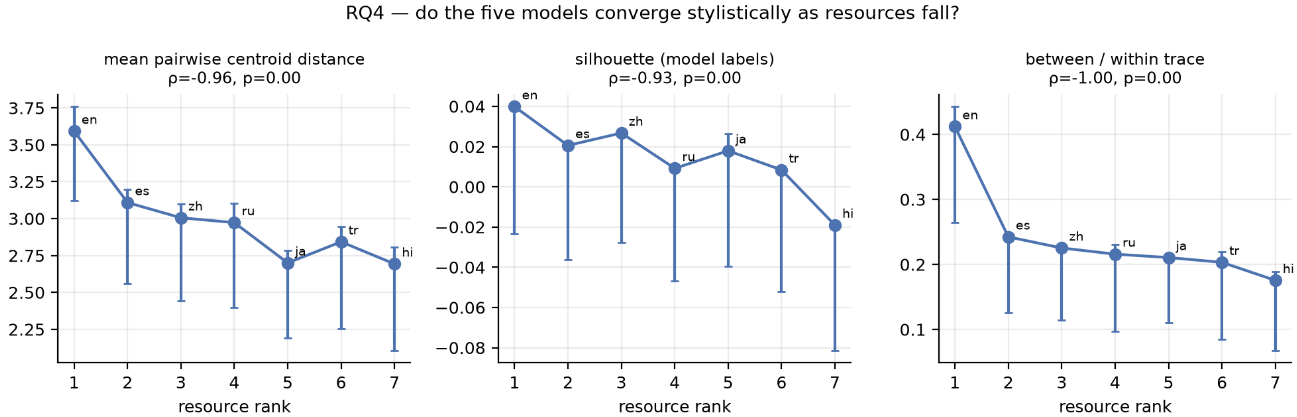

Figure 6. Three separation metrics against resource rank with reflected bootstrap CIs (1 000 resamples, stratified by model).

Table 6.1. Within-language model separation (features z-scored on the pooled language sample)

language	rank	centroid dist.	dist. CI lo	dist. CI hi	silhouette	sil. CI lo	sil. CI hi	between/within
English	1	3.592	3.123	3.758	0.040	-0.023	0.033	0.412
Spanish	2	3.109	2.559	3.196	0.021	-0.036	0.013	0.242
Chinese	3	3.006	2.441	3.098	0.027	-0.028	0.025	0.225
Russian	4	2.973	2.399	3.106	0.009	-0.047	0.002	0.216
Japanese	5	2.700	2.191	2.784	0.018	-0.039	0.026	0.211
Turkish	6	2.843	2.254	2.948	0.008	-0.052	0.007	0.203
Hindi	7	2.693	2.105	2.805	-0.019	-0.081	-0.019	0.175

Table 6.2. Trend tests

metric	Spearman ρ vs rank	p	English	Hindi
centroid dist	-0.96	0.000	3.592	2.693
silhouette	-0.93	0.003	0.040	-0.019
between within ratio	-1.00	0.000	0.412	0.175

Table 6.3. Pairwise centroid distance between models, within-language z-scored units

pair	en	es	zh	ru	ja	tr	hi
Claude Opus 4.7 – DeepSeek V4 Pro	2.35	2.73	1.67	2.92	1.67	2.42	2.57
Claude Opus 4.7 – GPT-5.5	3.49	3.38	2.50	3.64	3.18	3.79	3.58
Claude Opus 4.7 – Gemini 3.5 Flash	3.19	3.78	2.63	2.63	1.96	2.19	3.78
Claude Opus 4.7 – Grok 4.3	5.42	4.15	3.87	3.33	3.39	4.32	3.46
DeepSeek V4 Pro – GPT-5.5	2.14	2.08	2.81	2.43	2.63	2.12	1.41
DeepSeek V4 Pro – Gemini 3.5 Flash	2.09	2.11	2.25	2.60	1.81	1.28	2.02

pair	en	es	zh	ru	ja	tr	hi
DeepSeek V4 Pro – Grok 4.3	4.86	2.90	3.01	2.22	3.30	3.06	2.02
GPT-5.5 – Grok 4.3	5.69	3.75	3.99	3.83	3.80	3.32	2.69
Gemini 3.5 Flash – GPT-5.5	3.14	3.67	3.75	3.95	2.76	2.69	2.44
Gemini 3.5 Flash – Grok 4.3	3.55	2.55	3.58	2.18	2.49	3.25	2.96

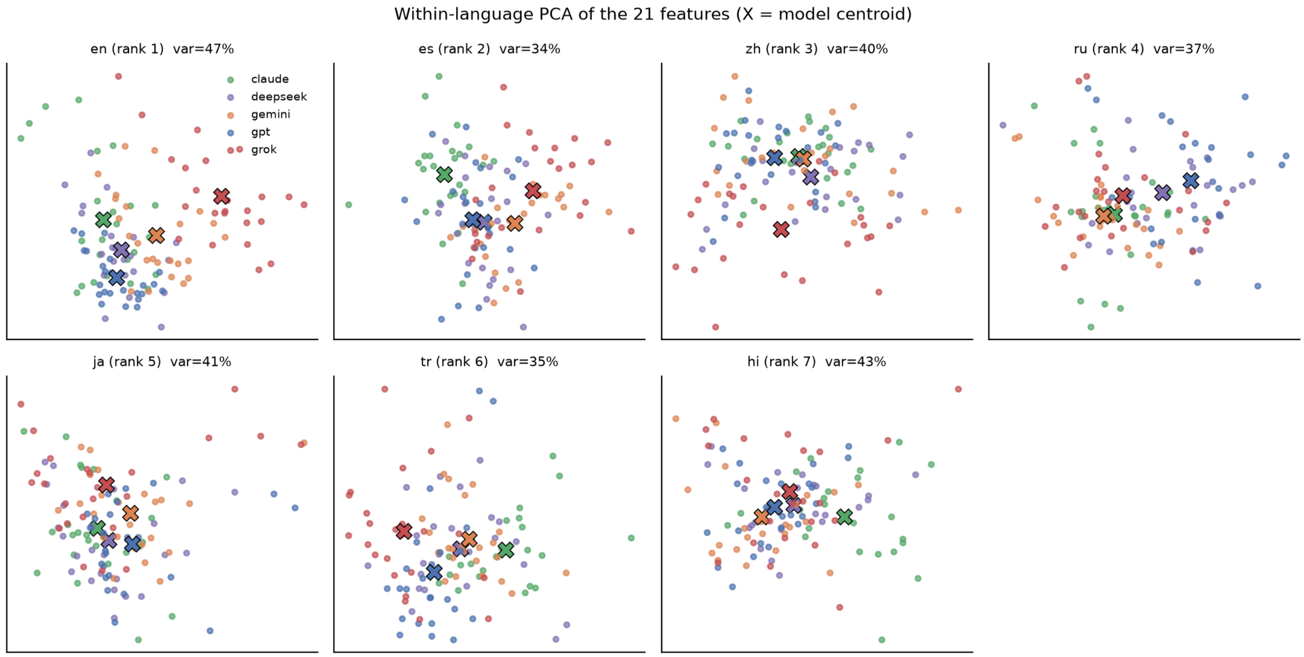

Figure 7. Within-language PCA of the 21 features; crosses mark model centroids.

Interpretation. All three separation metrics fall with rank, and all three Spearman correlations are at or near −1 (centroid distance $\rho = -0.96$, $p < 0.001$; silhouette $\rho = -0.93$, $p = 0.003$; between/within ratio $\rho = -1.00$). The between/within ratio more than halves from English (0.41) to Hindi (0.18); the silhouette is small everywhere and negative in Hindi, so no model forms a tight cluster in the low-resource languages. The pairwise table locates the convergence: the two largest English distances, GPT–Grok (5.7) and DeepSeek–Grok (4.9), collapse to 2.7 and 2.0 in Hindi, and Claude–Grok goes from 5.4 to 3.5. Grok's terse, sparsely punctuated English register is the outlier that disappears; the other four models were never far apart and stay that way. Convergence and sustained attribution accuracy are compatible because the classifier needs only a handful of features to separate five classes, while the centroid distance averages over all 21, most of which have converged. The practical reading for register: in low-resource languages the five models write more alike than they do in English, but not yet identically.

7 · RQ5 — does translation destroy the fingerprint?

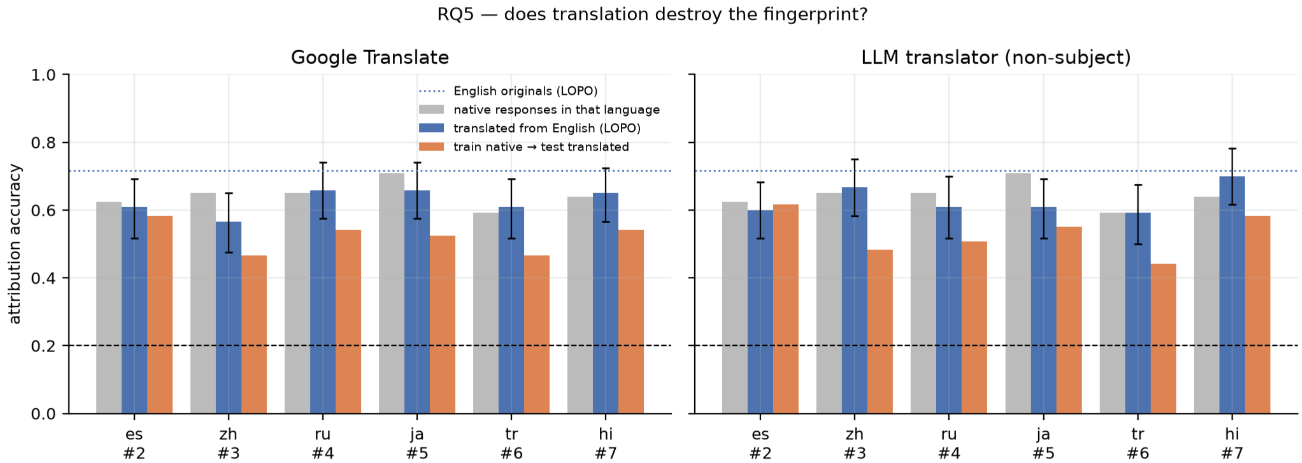

Figure 8. Attribution on translated text (blue) against native responses in the same language (grey), the English originals (dotted) and a native-trained classifier applied to the translations (orange).

Table 7.1. Attribution after translation (English originals: 0.717; every binomial test p < 0.001)

translator	language	n	translated, LOPO	CI lo	CI hi	native, same lang.	train native → test transl.	train English → test transl.
google	Spanish	120	0.608	0.517	0.692	0.625	0.583	0.717
google	Chinese	120	0.567	0.475	0.650	0.650	0.467	0.633
google	Russian	120	0.658	0.575	0.742	0.650	0.542	0.717
google	Japanese	120	0.658	0.575	0.742	0.708	0.525	0.575
google	Turkish	120	0.608	0.517	0.692	0.592	0.467	0.650
google	Hindi	120	0.650	0.567	0.725	0.639	0.542	0.692
llm	Spanish	120	0.600	0.516	0.683	0.625	0.617	0.750
llm	Chinese	120	0.667	0.583	0.750	0.650	0.483	0.625
llm	Russian	120	0.608	0.516	0.700	0.650	0.508	0.658
llm	Japanese	120	0.608	0.517	0.692	0.708	0.550	0.508
llm	Turkish	120	0.592	0.500	0.675	0.592	0.442	0.650
llm	Hindi	120	0.700	0.617	0.783	0.639	0.583	0.767

Table 7.2. Per-translator summary

quantity	google	llm
mean acc translated lopo	0.625	0.629
acc english originals	0.717	0.717
mean acc native same lang	0.644	0.644
mean acc train native test translated	0.521	0.531
mean acc train english test translated	0.664	0.660
mean feature rho	0.742	0.714
features rho above 0.5	16	16
mean eta2 model before	0.405	0.405
mean eta2 model after	0.321	0.299

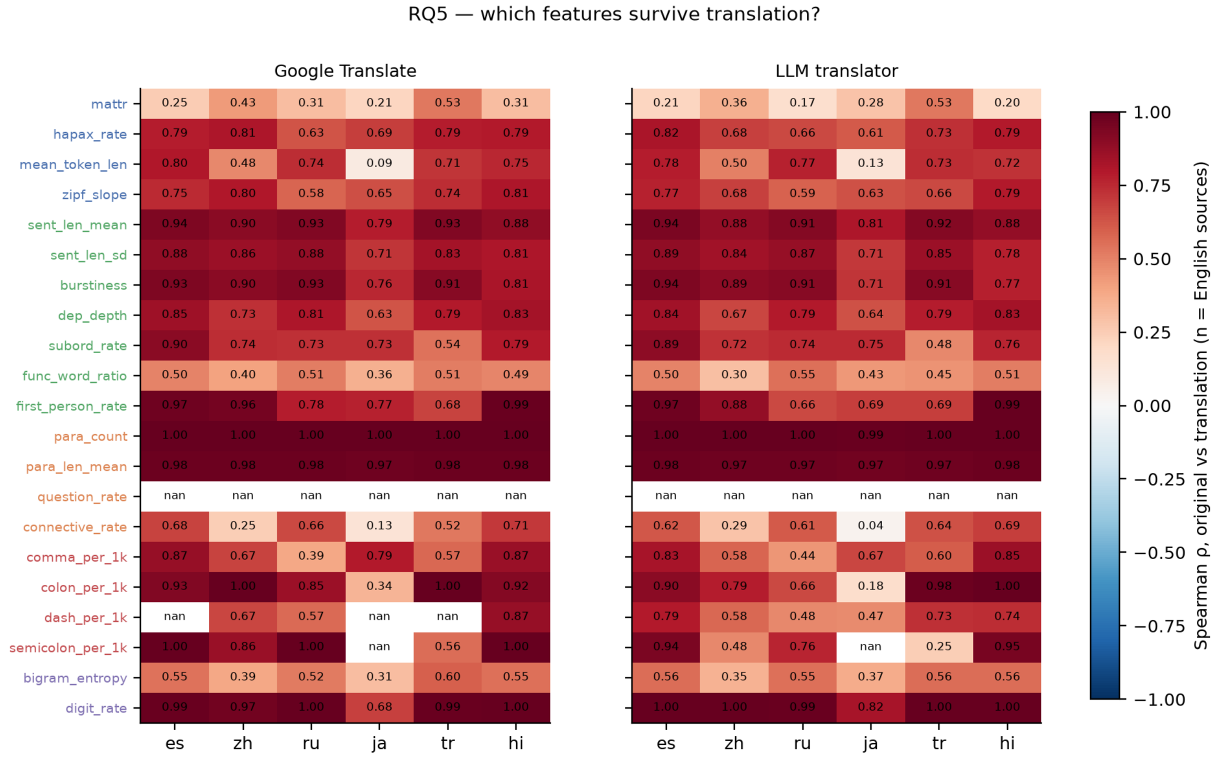

Figure 9. Which features survive translation: Spearman ρ between each English original and its translation, per target language.

Table 7.3. Mean Spearman ρ between original and translation per feature, averaged over the six target languages (question rate is undefined: the essays contain no questions)

feature	google	llm
para_count	1.00	1.00
para_len_mean	0.98	0.98
digit_rate	0.94	0.97
sent_len_mean	0.89	0.89
semicolon_per_1k	0.88	0.68
burstiness	0.87	0.86
first_person_rate	0.86	0.82
colon_per_1k	0.84	0.75
sent_len_sd	0.83	0.82
dep_depth	0.77	0.76
hapax_rate	0.75	0.72
subord_rate	0.74	0.72
zipf_slope	0.72	0.69
dash_per_1k	0.70	0.63
comma_per_1k	0.69	0.66
mean_token_len	0.60	0.61
connective_rate	0.49	0.48
bigram_entropy	0.48	0.49
func_word_ratio	0.46	0.46
mattr	0.34	0.29
question_rate		

Interpretation. Neither translator destroys the fingerprint. Attribution on translated text alone runs 0.57–0.70 (mean 0.625 for Google Translate, 0.629 for the LLM translator) against 0.717 on the English originals and a mean of 0.644 for *native* responses in the same languages; every CI excludes chance and every binomial test has $p < 0.001$. Translating a text costs about nine points of accuracy, roughly what moving from English to a native low-resource language costs, and no more.

The English fingerprint travels intact. A classifier trained on the English originals and applied to their translations scores 0.66 on average for both translators (0.77 for the LLM's Hindi), essentially the English accuracy, so what a translator outputs is still recognisably the source model's *English* style in another script. A classifier trained on the model's own native-language responses does worse on the translations (0.52–0.53): translated GPT resembles English GPT more than it resembles native Spanish GPT. This also answers the "which style does the translated text carry" question: the translator preserves the source's structural habits rather than re-styling the text into the model's native register for that language.

Feature survival explains why. Structural and rhythmic features come through almost unchanged (Spearman ρ between original and translation, averaged over six languages): paragraph count 1.00, paragraph length 0.98, digit rate 0.94–0.97, sentence length 0.89, burstiness 0.87, first-person rate 0.82–0.86, colon rate 0.75–0.84, sentence-length SD 0.83, dependency depth 0.77, subordination 0.72–0.74. Lexical features are rewritten by the target language: MATTR 0.29–0.34, function-word ratio 0.46, bigram entropy 0.48, connective rate 0.49. Sixteen of the twenty measurable features keep $\rho > 0.5$ under both translators. The share of feature variance explained by model falls from 0.40 in the English sources to 0.32 (Google) and 0.30 (LLM) after translation, a reduction, not an erasure. Question rate is undefined because these essays contain no questions.

Google Translate and the LLM translator behave almost identically. The one visible difference is punctuation: the LLM normalises semicolons and colons more (ρ 0.68 vs 0.88, 0.75 vs 0.84), consistent with an LLM lightly imposing its own conventions, yet its translations are, if anything, marginally *more* attributable. A free non-subject LLM therefore does not overwrite the source fingerprint with its own. For provenance work the practical reading is that translation is not an effective laundering step against interpretable features, and that a classifier trained in English can be applied across a translation boundary without retraining.

7b · RQ6 — LLM-as-judge attribution and self-recognition

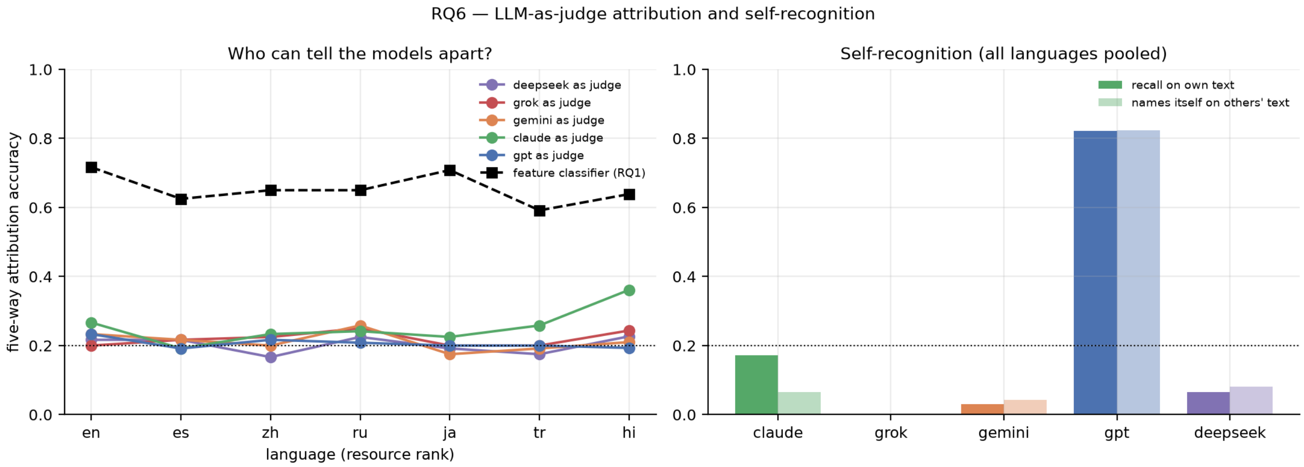

Figure 10. Left: five-way attribution accuracy per judge and language against the RQ1 feature classifier. Right: own-text recall versus the rate at which a judge names itself on others' text.

Table 7b.1. Per judge, all languages pooled (chance 0.20)

judge	n	accuracy	p vs chance	own recall	false-self rate	self-recognition p (Fisher)	self-claim rate	unparsed
Claude Opus 4.7	839	0.254	0.000	0.173	0.066	0.000	0.087	0.000
Grok 4.3	839	0.219	0.089	0.000	0.001	1.000	0.001	0.000
Gemini 3.5 Flash	839	0.212	0.201	0.030	0.043	0.844	0.041	0.001
GPT-5.5	839	0.206	0.340	0.821	0.824	0.583	0.824	0.000
DeepSeek V4 Pro	839	0.203	0.438	0.065	0.082	0.806	0.079	0.000

Table 7b.2. Five-way accuracy per language: each judge vs the feature classifier

language	Claude Opus 4.7	DeepSeek V4 Pro	Gemini 3.5 Flash	GPT-5.5	Grok 4.3	feature classifier (RQ1)
English	0.267	0.217	0.233	0.233	0.200	0.717
Spanish	0.192	0.217	0.217	0.192	0.217	0.625
Chinese	0.233	0.167	0.200	0.217	0.225	0.650
Russian	0.242	0.225	0.258	0.208	0.250	0.650
Japanese	0.225	0.192	0.175	0.200	0.200	0.708
Turkish	0.258	0.175	0.192	0.200	0.200	0.592
Hindi	0.361	0.227	0.210	0.193	0.244	0.639

Table 7b.3. Own-text recall per judge × language (self-recognition; chance 0.20)

judge	en	es	zh	ru	ja	tr	hi
Claude Opus 4.7	0.25	0.00	0.08	0.00	0.38	0.08	0.42
DeepSeek V4 Pro	0.00	0.00	0.12	0.08	0.12	0.04	0.08
Gemini 3.5 Flash	0.08	0.04	0.00	0.00	0.00	0.00	0.08
GPT-5.5	0.58	0.88	0.88	0.88	0.71	0.88	0.96
Grok 4.3	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Interpretation. Asked directly, the models cannot do what the feature classifier does. Four of the five judges sit at chance on every language: GPT-5.5 20.6%, Grok 21.9%, Gemini 21.2%, DeepSeek 20.3%, none significant, against 59–72% for the 21-feature classifier on the same texts. Their answers are not guesses spread evenly but fixed defaults: GPT names *itself* for 82% of all texts whether or not it wrote them (own-text recall 0.82, false-self rate 0.82, so no discrimination at all), while Grok, Gemini and DeepSeek almost never name themselves and call most texts "Claude" (Grok 681 of 839). Out of the box, attribution by an LLM is a prior about who writes essays, not a reading of the text.

Claude Opus 4.7 is the exception, as in the CompLLM result, and the exception is small but robust. Its overall accuracy is 25.4% (213/839, binomial $p < 0.001$), and it is the only judge whose own-text recall (17.3%) exceeds the rate at which it names itself on others' text (6.6%; one-sided Fisher $p < 0.001$). Its self-claims are also unusually precise: when Claude says "Claude" it is right 40% of the time against a 20% base rate, and when it says "Gemini" it is right 81% of the time; it simply under-calls both, defaulting to "GPT" for 614 of 839 texts. Its above-chance accuracy therefore comes from cautious discrimination rather than from self-recognition alone (accuracy on others' text 0.27, on its own 0.17).

The self-recognition signal does not fade with resource level; if anything it is strongest where the resource hypothesis predicted it should vanish. Own-text recall versus false-self rate is significant in English (25% vs 3%, $p = 0.002$), Japanese (38% vs 16%, $p = 0.02$) and Hindi (42% vs 13%, $p = 0.003$), where Claude's overall accuracy also peaks at 36.1% ($p < 0.001$). In Spanish and Russian Claude never names itself at all (0 of 24 own texts), so the signal is absent there rather than reversed. The correlation of own-recall with rank is positive and non-significant ($\rho = 0.45$, $p = 0.31$): there is no gradient, and the low-resource end is where Claude is most willing to claim its own writing.

Two caveats. First, the effect is modest in absolute terms; a 25% judge is far from a usable attributor, and the paper's practical claim should be that interpretable features beat every LLM judge by 35–50 points in every language, not that Claude can identify itself reliably. Second, the judge condition disabled hidden reasoning for the four models that allow it, so this is the instant-answer setting; Gemini kept low-effort reasoning because it cannot be disabled, and still sat at chance, which argues against reasoning being what the others were missing.

8 · Limitations

- **One prompt family.** Twelve persuasive-essay schemas with a fixed three-point structure. The paragraph-count and connective features that carry much of the fingerprint are partly a property of how each model handles that instruction; other genres may fingerprint differently.
- **Length is entangled with style.** Models differ systematically in length and the linear classifier uses it. The length-residualised run keeps attribution well above chance (non-linearly), but a deployment classifier should treat length as a feature to justify, not a given.
- **Resource rank is ordinal.** The gradient test has seven points; the cell-level GLM adds power but inherits the same coarse covariate.
- **Reasoning budgets.** Ten DeepSeek cells overflowed a 4 000-token budget with hidden reasoning and were re-collected at 16 000; the override is recorded per response. One Grok response to a Hindi prompt came back in English and was excluded. 839 of 840 cells are used.
- **Two generations per cell.** Within-cell variance is estimated from $n = 2$; the leave-one-prompt-out design prevents leakage but confidence intervals are correspondingly wide (±7–9 points).
- **Automated prompt review.** Prompt fidelity was checked by a model, not a person, in all seven languages. The review and every regenerated wording are versioned for audit.

- **Judge condition.** RQ6 disables hidden reasoning for the four judges that allow it (instant-answer condition); Gemini keeps low-effort reasoning. A reasoning-on condition was piloted and abandoned for cost; it may raise judge accuracy and should be tested before claiming that LLM judges *cannot* attribute.
- **Snapshot.** Model strings are dated snapshots served through OpenRouter in September 2026; fingerprints will drift with model updates.

9 · Reproduction

Code, prompts, every raw response, translation, feature table and result file are versioned at github.com/adrian-erlikhman/LangLLM. The pipeline is a handful of resumable commands:

```
python -m langllm.prompts; python -m langllm.review; python -m langllm.refine
python -m langllm.collect; python -m langllm.validate; python -m langllm.features
python -m langllm.analysis; python -m langllm.figures
python -m langllm.translate; python -m langllm.features --translated
python -m langllm.analysis_translation; python -m langllm.figures --rq5
python -m langllm.report; python -m langllm.artifact; python -m langllm.pdf_report
```

Total API spend for the study, including prompt writing, review, collection and 1 440 translations, was under fifteen US dollars.

Appendix

Table A.1. Model × language cells with any exclusion (final data, after the DeepSeek re-collection)

model_key	lang	n	wrong_language_rate	refusal_rate	truncated_rate	too_short_rate	too_long_rate	keep_rate
Grok 4.3	Hindi	24	0.04	0.00	0.00	0.00	0.00	0.96

Table A.2. One-way ANOVA F of each feature by model, within language (higher = more separable by that feature alone)

feature	en	es	zh	ru	ja	tr	hi	mean F
comma_per_1k	12.5	25.7	18.8	21.2	48.6	1.0	8.6	19.5
zipf_slope	56.3	7.1	21.9	9.9	17.7	9.2	10.8	19.0
hapax_rate	48.1	6.8	23.4	10.9	14.0	10.3	9.8	17.6
bigram_entropy	2.2	9.3	39.0	5.2	27.7	12.4	7.1	14.7
para_count	8.5	13.4	6.1	16.9	11.5	18.7	16.9	13.1
colon_per_1k	12.5	31.5	1.9	23.9	2.0	10.0	2.6	12.1
mattr	28.2	10.1	14.6	2.8	1.0	2.1	2.1	8.7
para_len_mean	7.7	9.0	1.1	20.5	2.4	6.7	4.9	7.5
mean_token_len	31.9	3.1	10.3	3.2	0.4	2.2	0.7	7.4
connective_rate	2.8	5.8	11.9	7.7	2.7	14.3	5.7	7.3
func_word_ratio	18.1	6.0	9.5	3.7	0.7	5.3	0.9	6.3
burstiness	12.9	9.8	0.7	3.5	4.0	3.0	9.7	6.2
dep_depth	19.9	6.3	2.0	0.8	2.5	6.5	5.2	6.2
subord_rate	16.1	1.8	1.2	4.8	1.8	7.4	0.2	4.8
sent_len_mean	9.1	6.1	1.1	1.6	3.4	6.2	5.5	4.7
dash_per_1k	3.7	2.5	3.1	5.8		2.9	9.1	4.5
sent_len_sd	7.4	4.8	1.4	3.0	5.5	4.1	5.3	4.5
first_person_rate	4.0	6.9	3.7	1.0	4.9	2.1	4.9	3.9

feature	en	es	zh	ru	ja	tr	hi	mean F
digit_rate	1.7	1.5	1.1	5.3	1.3	5.4	2.2	2.7
semicolon_per_1k	4.0	1.6	3.0	1.7		2.2	2.2	2.5
question_rate		0.8	1.0	0.9		1.8	0.5	1.0